# DENIZHAN TOPRAK

## GAME DEVELOPER

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### SUMMARY

Computer Engineer specialized in Game Development. I have worked with multiple companies and strengthened both my technical and collaborative skills. I enjoy creating tools and robust system designs that make projects more enjoyable, efficient, and scalable.

### PROFESSIONAL EXPERIENCE

**Gameplay Programmer - Tigerroll Studios** *United Kingdom - Remote* **Dec 2025 - Active**

- Developed **core gameplay systems** for **MMORPG** experiences, including player-to-player and NPC interaction frameworks using **Unreal Engine 5**'s **Gameplay Ability System (GAS)**.
- Led implementation of **cross-platform**, gamepad-friendly UI architecture using Unreal Engine's **Common UI** framework.
- **Coordinated** between design and engineering teams by organizing **Figma** workflows and establishing clear pipelines for translating UI designs into in-game interactive UI systems.
- Designed and implemented complex NPC **gameplay AI** using **State Trees**, enabling dynamic behaviors such as *combat*, *dialogue systems*, and *NPC-to-NPC interactions*.

**Archinord Software - C++ Developer** *Turkiye - Office* **Sep 2024 - July 2025**

- Using **Unreal Engine 5** I led the development of multiple architecture visualization projects.
- Created **Engine Tools** to get live data from **AWS** servers to display live information about the residences (Such as *price, media, and other metadata) to display to the product users and developers*.
- Created project **DLC - Patch System** to update our client's projects using **AWS Cloud Systems** and **Unreal Engine 5** to reduce development time and increase user feedback.

**Retter Business Services - Software Developer** *Turkiye - Remote* **July 2021 - July 2023**

- Created **NodeJS backend** methods for retail web projects that are used in *login*, *order*, cart and payment that were run on **AWS**'s several web tools.
- Participated in various projects from the start, documenting and addressing issues. (including *Pınaronline*, *Jotun Usta Fırçalar*, *Koçailem*, *Unishop*.)
- Used **Jira** as the **project management tool** to create, manage, follow and test project tasks for our developers.
- Developed **generic automated backend tests** mainly in **N8N**, enhancing bug detection and project stability.
- **Created documentation** for the *Retter* for their custom web development tools.

**Indie Game Project - Lead Developer** **Dec 2023 - Active**

- Developed a full-scale game project, building gameplay systems, AI, UI, tools, and core mechanics.
- Designed modular and scalable architecture using **C++** and **Unreal Engine** frameworks.
- Implemented custom systems (interaction, combat, inventory, tools) later extended into standalone plugins.
- Managed all aspects of development including planning, project management, prototyping, optimization, and technical documentation.
- Created **multiplayer** beta builds and made unit tests.
- Created a cross-platform **UI Framework** that is build on top of the Epic's **Common UI.**

## SKILLS

**Jan 2023 - Jun 2023**

| | | |
|---|---|---|
| Unreal Engine 5 | Animation - Cascadeur | SCM - Git, PlasticSCM, Perforce |
| Tools Development | 3D - Blender | Visual Studio, Rider, VsCode |
| User Interface | C++, C#, Java, Python, Js | Working Remote |
| Shaders & VFX | AWS | Agile Development |

## EDUCATION

**Bachelor of Computer Engineering** **Sep 2019 - July 2023**
Marmara University

## PROJECTS

***Unreal Engine C++ Plugins***

Some of my plugins are listed with their descriptions. Further information is in the documents.

- **Stage Interaction System:**
  - Stage based approach for gameplay interactions, where each interaction state can be handled independently and modularly in multiplayer games using **State Trees.**
- **Ability Queue Task System:**
  - Using the full extend of **Gameplay Ability System** by Epic, with additional, non-destructive, customizable **Input Queue** system that enables fluid action transitions for any type of game.
- **Modular Inventory System:**
  - Object based replicated inventory system where each item has the potential to hold other items as containers in a grid based system. Items use **Data Registry, GameplayTags, Instanced Data Fragments** in order to make modular and scalable systems.
- **Easy Skeletal Mesh Merge Async:**
  - Merges skeletal meshes to one at runtime in networked environments to optimize our players, NPC's, weapons etc. further.
- **Easy Ability Set System:**
  - Extension to **Gameplay Ability System.** By handling gameplay ability granting-clearing in bulk it gives desired friendly UI to studios with extra features: Auto Input-Ability binding, Effect Sets etc.
- **Dynamic Input System:**
  - Flexible, runtime input management system for Unreal Engine that allows developers to dynamically bind, modify, and manage input mappings during gameplay based on game context and displaying their metada for the players (*e.g. LMB to Attack, Space to Jump.*)

**Software Development Projects**

Some of my software development projects are listed here with their links.

- **Complaint_tr** :
  - Using **Microsoft AspNetCore** we have developed a working example complain system on web.

## ADDITIONAL INFORMATION

- **Languages:** English, Turkish.
- **Programming Languages:** C++, NodeJS, Javascript, Java, C#, Python
- **Student Clubs:** Marmara University Game Development - Founder/Chairman